Module 6
Biomechanical Assessments

Biomechanics is one of the most important aspects of running because of its direct impact on performance. Biomechanics plays a large role regarding the efficiency of a runner. So what is biomechanics? The textbook definition of biomechanics is as follows (360):

> ***Biomechanics is the application of mechanical principles in the study of living organisms.***

Simply put, the application of biomechanics in regard to running is to examine, assess, and modify the physical position and/or form of a runner to allow the individual to perform as efficiently as possible. It is your job as a coach to get as much speed and power out of your client with the lowest energy expenditure.

This module discusses biomechanical assessments in regard to movement patterns, increasing running economy, posture, and potential reasons for biomechanical issues.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Biomechanical terminology
- Lever classes
- Effect of muscle synergy on biomechanics
- Kinetic chain
- Static and dynamic postural screenings
- When to refer clients to specialists

❖ TERMINOLOGY

Pronation: medial (inward) rotation of a body part

Supination: external rotation of a body part

Force: a pushing or pulling motion; the product of mass and acceleration

Friction: a force acting on the area of contact between two surfaces in the direction opposite the motion or motion tendency (361)

Drag: resistance created between the difference in pressure between the front and back of an object moving through air or fluid

Torque: rotational effect of force

Shear: force that is parallel to the surface

Open-Chain Exercise: exercise where the hand or foot is free to move and not in a fixed position

- Ex: Dumbbell Press: Hand moves freely in the air

Closed-Chain Exercise: exercise where the hand or foot is in a fixed position and cannot move

- Ex: Push-Up: Hand is in a fixed position on the ground

❖ INTRODUCTION

While many coaching programs focus solely on the cardiovascular aspect of training, if biomechanics are not addressed, clients will not be able to perform to their potential and their chance of getting injured likely increases. For these reasons, paying attention to and addressing a client's form is of critical importance.

Biomechanics and proper form provide the foundation on which a training program is constructed.

Biomechanics can be broken down into two categories, *static* and *dynamic*.

> **Static:** Refers to the analysis of systems that are either at rest or in constant motion (moving at a constant velocity). For the purpose of this certification, static biomechanics refers to a system at rest (i.e., no motion).
>
> **Dynamic:** Refers to the analysis of systems in which movement and acceleration are present.

Dynamic biomechanics can be broken down even further into two areas: ***kinematics*** and ***kinetics***. Kinematics describes the appearance of motion, whereas kinetics studies the forces associated with motion. The area that concerns the study of human movement is ***kinesiology***.

As muscle imbalances can affect the biomechanics of a runner, when assessing a runner biomechanically, it is important to look beyond the surface to see what may be the reason for a biomechanical inefficiency.

Many factors come into play regarding performance. For example, a particular running form might be the most biomechanically efficient. However, it might not be sustainable because of the client's lack of flexibility. This demonstrates how many factors influence the efficiency and optimal performance of a runner (running economy).

A portion of this module discusses postural screenings and potential causes for postural issues. These screenings are primarily noted to inform you in regard to what is considered correct and incorrect posture and form.

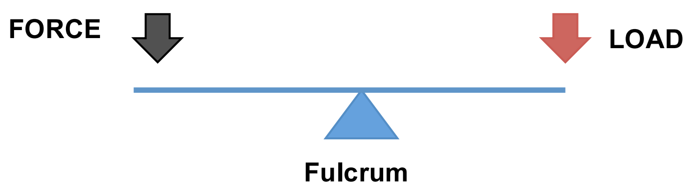
Being informed in this area does not necessarily equate to being qualified to address these issues. The cause of a postural or form issue may not be initially evident because of how the kinetic chain affects the body. While this certification is quite robust and discusses potential reasons for biomechanical and postural/form deviations, these areas are discussed primarily for the purpose of education versus being a catalyst for action on your part.

It is **always** the correct choice to refer clients to a clinical professional such as a physical therapist if you believe they have a postural issue.

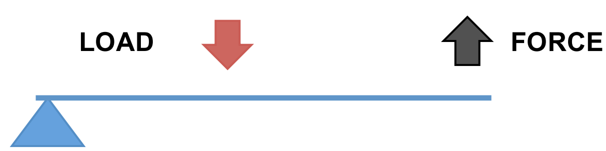
THREE LEVER CLASSES

At the core of biomechanics are levers. It is important to understand how levers work to fully appreciate the efficiency of the body in regard to positions and movements. A lever in its simplest form is an object that utilizes a pivot point (fulcrum) to gain a mechanical advantage when moving a load. Following are the three classes of levers found in the body (56).

Class 1: fulcrum is located between the applied force and the load (e.g., upward tilting of the head)



Class 2: the load is located between the applied force and fulcrum (e.g., jaw opening)



Class 3: applied force is located between the force and load; this classification of lever is the most common in the body (e.g., bicep curl)

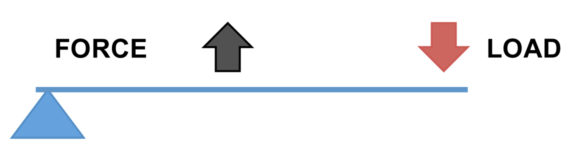


Figure 6.1 Lever Classes

Biomechanically speaking, the body consists of a bunch of levers held together by muscles and connective tissue. Therefore, when assessing or enhancing a runner's efficiency, the place to start is the mechanics of the body. The goal of a runner is the following:

> ***Establish and maintain form that minimizes energy expenditure and drag while maximizing structural support and power output.***

Many individuals look to replicate the posture and form used by elite and professional athletes. This is because many amateur runners associate the training methodologies and biomechanics/form of professionals with sport success. This can be a slippery slope. Assuming an elite athlete has correct form, there is a good chance your client will not be able to replicate it. Even if your client can replicate the form, unless the position is sustainable it should not be put into practice. Additionally, just because an athlete is a professional does not necessarily mean the individual has correct form and mechanics. As a coach, it is your responsibility to explain the reasoning behind your training program and especially areas related to form and biomechanics.

It is helpful to think of biomechanics as science-based rather than sport-based. Your ability to assess and make adjustments to your client's form needs to be based in science and not exclusively on knowledge of an individual sport.

LEARNING AND RELEARNING PROPER FORM

Your clients will fall into three main areas regarding learning correct form:

1. Teaching proper form to clients who have never run.
2. Teaching proper form to clients who are currently running with incorrect form.
3. Your client currently has the correct form.

Let's start by addressing #3. If your client has proper form, you can breathe a sigh of relief and move on. However, you should be aware that this is extremely rare, as even clients with seemingly perfect form typically have small areas that can be improved upon.

Have you ever heard the phrase *"It's hard to unlearn a learned behavior"*? According to Dr. Richard Schmidt, a motor behavior expert, it takes between 300 and 500 repetitions to "learn" a movement but between 3,000 and 5,000 repetitions to change an existing motor program (144). Whether or not these numbers are spot-on, it clearly demonstrates that it is much easier to learn a motor program for the first time than to unlearn an established motor program and then relearn the correct motor program. This relates directly to muscle memory, as discussed previously.

Two things are clear: First, whether or not a client is learning a motor program (e.g., correct form) for the first time or relearning a motor program, it takes time to develop proper form. Second, if a client's form is incorrect but the person has a goal event coming up soon, unless your client is at a physical risk by maintaining current form, it is advised to focus on changing form after the goal event. By attempting to modify a client's form too quickly, performance will likely be negatively impacted. As an example, if a basketball player is a 90 percent free throw shooter but has poor form, if you were to try to correct the player's form midseason, performance (free throw percentage) is almost guaranteed to decrease in the short term.

INJURY

Within the sport of running, injuries can result from many things, including muscle imbalances, incorrect form, and poor mechanics. These areas are typically interrelated. However, incorrect form and poor mechanics/posture are not always predictive of pain or injury.

MUSCLE SYNERGY AFFECTS MECHANICS

As discussed in *Module 3*, the body acts as a kinetic chain, meaning a biomechanical deficiency in one area will likely affect other areas of the body.

The length of a muscle is relevant because in order for correct posture to be maintained, muscles need to function in equilibrium. If this equilibrium is not maintained, muscle imbalances will result and are often visible in the form of incorrect posture and incorrect movement patterns. Two terms that relate to muscle length will be used throughout the certification.

- **Shortened**
    - *Similar Terms:* overactive, tight, short

- **Lengthened**
    - *Similar Terms:* underactive, weak, long, inhibited

Regardless of the reason for a postural or biomechanical issue, there is a good chance that the issue has existed for a while. Assuming this is the case, the issue most likely will not be fixed overnight. Quite often, the process of correcting a biomechanical issue involves strengthening weak and underactive muscles and relaxing tight muscles to restore muscle balance and proper function. If your client has long-standing and/or significant biomechanical issues, you need to refer the person to an appropriate clinical professional. Physical therapists have advanced training in orthopedics and often use modalities and treatment techniques that coaches are not qualified to utilize.

No individual is perfect in regard to posture, symmetry, or biomechanics. For example, an individual might have one shoulder that is slightly higher than the other. Does this individual have a biomechanical issue? Yes. Is it a biomechanical issue that presents a problem in regard to sports performance or pain? Perhaps.

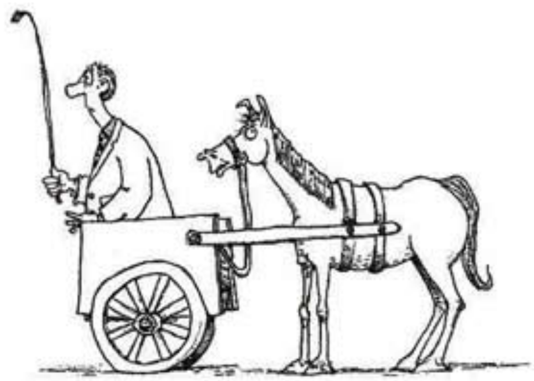
While it is always the correct course of action to refer your clients to a specialist if you feel they have an issue that would benefit from a specialist's intervention, you **must** refer them to a specialist in the following cases:

- Client experiencing pain that is not eliminated by form or shoe modifications
- Substantial biomechanical deviation or muscle imbalance (e.g., easily observable)
- Biomechanical deviation or muscle imbalance resulting from an injury

Your understanding of biomechanics is important, as you need to be able to communicate with your client as well as the therapist regarding potential issues and corrective measures.

As noted in the *Muscular System* module, restricted fascia can negatively influence and alter proper body mechanics.

PUTTING THE CART BEFORE THE HORSE



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Many coaches feel that the ability of a client to run with proper form is solely a function of effective cueing, the client learning how to do the exercise, or learning particular movements. While these are components of a client being able to perform a movement with correct form, it is only one aspect.

If your client is biomechanically unable to perform a movement because of muscle imbalances or a breakdown in the kinetic chain, no amount of cueing or effort on the part of the client will enable the person to perform the correct movements or have a proper body position.

This is akin to an individual who is able to do only addition, subtraction, and multiplication and then asking the person to do calculus. It is not going to go well.

As a coach, you can ask your clients to perform a movement or position their body in a particular way only if they are biomechanically capable. Until this occurs, there must not be an expectation on your part that a particular element of proper form is attainable.

An example of this would be a client whose upper body leans substantially forward at the hips when running. While there are many possible reasons for this, the most likely reason is a muscle(s) imbalance. Assuming this is the case, until the imbalances are addressed you cannot expect that just by explaining the proper form, the client will be able to adjust and maintain the “correct” form.

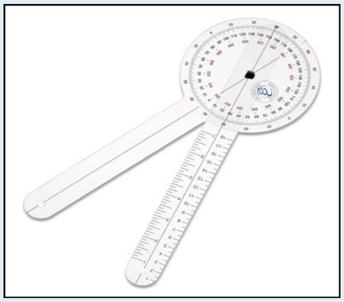
PERFECTION IS NOT ATTAINABLE, NORMAL, OR NECESSARY

While it is important to assess a client regarding postural and biomechanical issues, it needs to be stated that it is normal for individuals to have postural asymmetries, muscle imbalances, and biomechanical deviations (452). As noted earlier, these issues are not necessarily predictive of pain or dysfunction (452).

In a research article by the *Center for Professional Development In Osteopathy and Manual Therapy*, it is theorized that the body has the capacity to adapt to biomechanical inefficiencies to the extent that normal functioning is not reduced and pain is not present (452). Therefore, to a certain degree, the body has the ability to adapt to inefficiencies without any negative impact on the individual.

❖ ASSESSING THE CLIENT

What Is a Goniometer?



A goniometer is an instrument that is used to measure angles of the body. In clinical settings, it is used to assess range of motion. A goniometer consists of a fixed arm and an arm that rotates around an axis.

HOW TO USE A GONIOMETER

Position the axis of the goniometer at the fulcrum of a joint. Line up the goniometer arms with the body parts that make up the angle being assessed.

IS IT ACCURATE?

A 2009 study examined the validity of using a manual goniometer to assess the knee angle and found that in comparison with a computer navigational system, the goniometer was less accurate than the computer system (444). It can likely be assumed that the skill/experience of a tester has a lot to do with the accuracy. Additionally, short goniometer arms may also decrease the accuracy because of the arms' not reaching the landmarks being assessed.

DIFFERENCE BETWEEN JOINT ANGLE AND FLEXION/EXTENSION

The word *angle* is often used to describe flexion/extension. For example, when the foot hits the ground while running, the knee angle may be 145 degrees. However, this angle is often referred to as 35 degrees (180 minus 145). In this case, 35 degrees corresponds to knee flexion, not the knee angle.

CONCLUSION

Use of a goniometer is most useful and practical in clinical settings to assess ROM pre-/post-injury or during rehabilitation.

Before we go any farther in this section, it is extremely important to note that your findings during screenings **DO NOT** represent a diagnosis of any sort. You are a running coach, not a physician or physical therapist. This cannot be stressed enough.

Assuming you have noted a potential mechanical/postural issue during one of the following screens, the proper verbiage to a client must not declare a definitive issue, but rather note that the altered posture/movement pattern “suggests” that the pattern “might be” negatively affected by a particular musculoskeletal issue.

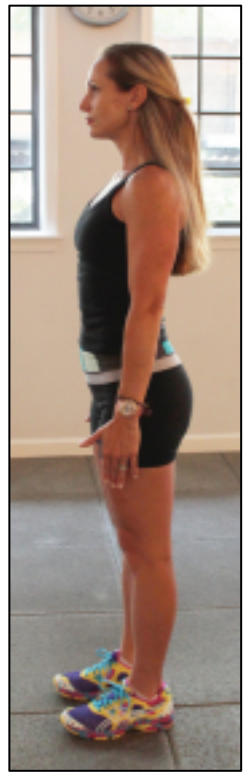
Why not use a definitive statement? You cannot legally diagnose a musculoskeletal issue, and more to the point, there are typically multiple issues that affect an altered movement pattern. As earlier discussed, this is why UESCA strongly recommends referring clients who present with pain, substantial biomechanical issues, or biomechanical issues that result from injury to a specialist such as a physical therapist.

This issue is discussed in greater detail in the *Refer Out* section of this module.

STATIC POSTURAL SCREENINGS

By walking around your client in a standing position, you can visually assess the individual from the following points of view: anterior, posterior, and lateral. Any major postural issues that are evident at first glance should be referred to a physical therapist.

Anterior View



Lateral View

Figure 6.2 Static Postural Screenings

Anterior Screen

Shoulder Level: The shoulders should be in an even, horizontal plane and not elevated.

- Individuals with scoliosis and/or a tilted or rotated pelvis often present with uneven shoulders.

- If the shoulders are grossly uneven, the head typically tilts laterally to the side of the higher shoulder in an attempt to maintain equilibrium.

Hip Level: The hips should be in an even, horizontal plane and not rotated in the frontal plane.

- **Leg Length Discrepancy (LLD)**
  Often discussed in regard to running, a leg length discrepancy occurs when one leg is shorter than the other. An LLD can cause a host of issues such as back and knee pain, just to name a few. There are two types of LLDs: *structural* and *functional*.

  A structural leg discrepancy is when one leg is truly a different length than the other leg. A functional leg length discrepancy is when one leg appears to be a different length than the other because of injury, muscle imbalances, or biomechanical issues. Functional LLDs are much more common than structural LLDs.

  Below are common traits of an individual with an LLD:

  - Tilted pelvis in the frontal plane
  - Uneven shoulders in the frontal plane
  - Standing with one knee flexed
  - Back pain

  If you suspect your client has an LLD, you need to refer the individual to a physical therapist who can assess whether or not the client has an LLD and if it is structural or functional.

Legs: From the hips to the ankles, the legs should visually be in a near straight line.

- Knees that significantly bow in (valgus) or out (varus) need to be assessed by a physical therapist.

Lateral Screen

Vertical Body Alignment: Any excessive fore/aft positioning of the head, neck, or spine should be referred to a specialist.

Hips: Hips that are rotated in the transverse plane should be referred to a specialist. Unless hip rotation is substantial, this is very difficult to assess visually. If your client always stands with the same leg in front, this could be indicative of a rotated pelvis in the transverse plane.

Shoulders: Observe if the shoulders are excessively internally/externally rotated. Of the two, internally rotated shoulders are a more common postural issue and are often caused and/or exacerbated by long durations spent hunched over a desk.

DYNAMIC POSTURAL SCREENINGS

Figure 6.3 Overhead Squat Screen-Correct Form (Lateral View)

This section discusses postural assessments that are specific to running as well as general posture.

The goal of the certification is not to list every assessment or postural deviation. Rather, the purpose of this section is to discuss several screenings that cover a wide range of postural issues that are prevalent among runners.

Overhead Squat Screen

The *overhead squat screen* (OSS) is one of the best all-around functional tests to assess a client's strength, flexibility, and neuromuscular efficiency (80). To perform the OSS, have your client stand with legs shoulder width apart with arms extended directly over the head with elbows locked. After this position has been achieved, instruct the client to squat with body weight over the heels. An effective way to verbally cue the squat is to tell the client to squat down and back like sitting down on a low chair (i.e., stick the butt out). You also want to cue the client to keep the head facing forward, back as straight as possible, chest forward, and maintain the overhead arm position with the elbows locked.

When assessing the client, you need to move around to view the person from all angles, as many postural deviations can be seen only from certain angles. As the individual is performing the screening, take notes on your observations.

Knees

Knees Inward **Knees Forward**

Figure 6.4 Overhead Squat Screen – Incorrect knee movements (frontal/lateral views)

Correct Form: knees stay in the same vertically aligned position
Postural Issue: knees buckle inward when squatting, especially after fatigue sets in
Potential Causes: this may indicate underactive glute maximus/medius muscles, overactive hip adductors; inward knee movement can also be exhibited because of foot pronation and rotation of the pelvis in the transverse plane
Action: STRENGTHEN: glute maximus/medius
STRETCH: hip adductors

Correct Form: knees stay in the same vertically aligned position
Postural Issue: knees move forward in the sagittal plane
Potential Causes: this may indicate tight hamstrings, calves
Action: STRETCH: hamstrings, gastrocnemius, soleus

Lower Back

Figure 6.5 Overhead Squat Screen-Low back flexion

Correct Form: erect upper body (i.e., torso)

Postural Issue: once the squat is initiated, the spine begins to flex forward (rounds)

Potential Causes: this is often indicative of poor hamstring flexibility; once the hamstrings cannot stretch any farther, the lower back compensates by flexing; spinal flexion can also be exacerbated by weak spinal extensor muscles (erector spinae)

Action: STRETCH: hamstrings, abdominals
STRENGTHEN: erector spinae, multifidus, glute maximus

Arms / Torso

Figure 6.6 Overhead Squat Screen – Forward torso lean/arms fall

ARMS

Correct Form: arms stay in the same line as torso (when looking at the body laterally)
Postural Issue: arms fall forward in the sagittal plane and/or elbows flex
Potential Causes: possibly indicative of tight latissimus dorsi and pectoralis major; under active lower trapezius, rhomboids

Action: STRETCH: latissimus dorsi, pectoralis major
STRENGTHEN: lower trapezius, rhomboids

TORSO

Correct Form: erect upper body (i.e., torso)
Postural Issue: upper body tilts substantially forward at the pelvis
Potential Causes: overactive rectus abdominis, hip flexors, and calf muscles. If the calf muscles are tight, the ankle joint cannot flex enough,

resulting in the upper body bending forward to maintain balance. Underactive spinal extension muscles.

Action: STRETCH: rectus abdominis, gastrocnemius, soleus, hip flexors
STRENGTHEN: erector spinae, glute maximus

HEELS

Figure 6.7 Overhead Squat Screen – Heels rise off ground

Correct Form: as the individual squats, the heels stay flat against the floor

Postural Issue: the heels rise off the ground so that the body weight is primarily distributed through the balls of the feet and toes

Potential Causes: tight calves

Action: STRETCH: soleus
STRENGTHEN: tibialis anterior

WHEN TO REFER OUT

As noted earlier, postural screenings are not for the purpose of diagnosis. Rather, the purpose of the screenings is to identify breakdowns in form that could be suggestive of muscle imbalances.

While each screen lists potential causes of the postural deviation(s), as the body works in a kinetic chain, the origin of the postural issue could be somewhere other than the noted muscle(s).

Therefore, if your client is in pain or presents with a postural issue, it is strongly advised to refer the person to a specialist such as a physical therapist.

If you refer your client to a specialist, you may be brought into the treatment conversation with the client and specialist. This is a primary reason why this certification discusses postural and form issues. It is important to have a base knowledge of how the body functions so that you can discuss with a client and specialist on an informed and educated level.

Prevention vs. Intervention

Most people associate physical therapy with rehabilitation from an injury. While this is often the case, seeking out a physical therapist to help correct postural and mobility issues can be advantageous to help prevent injury. It should be noted that using a physical therapist who has a background of working with endurance sports athletes, while not necessary, is often advantageous.

For the nonathletic population, poor body alignment is not necessarily predictive of pain and injury (362, 363). This can also be true of endurance athletes. However, because of long training durations and the repetitious nature of distance running, it makes sense that there would be a higher correlation between poor body alignment/mobility and pain/injury.

Referring your client to a physical therapist is valuable for many reasons:

Decreased Liability

- Client has a serious orthopedic issue that is not fully healed, even if the client believes he or she is fine
- Client has a serious undiagnosed orthopedic issue

In both of these cases, you should likely cease working with this client until the person is discharged by the physical therapist, or at least directed by the therapist in regard to what activities and exercises the individual should and should not be doing. If you work with this client without referring him or her to a physical therapist, you will be at an increased liability risk if an injury occurs while you are working with the individual.

Education
A physical therapist can educate you on pertinent exercises/stretches for your client and the rationale behind them.

Time Sensitivity
A physical therapist will be able to identify postural and mobility issues quickly and therefore be able to devise a plan to get a client functioning at a higher level as soon as possible.

Ongoing Resource
Throughout the process of working with a client, having a professional resource (e.g., a physical therapist) whom you or the client can contact if need be is of substantial value.

Working with a physical therapist should be viewed not as two individuals with differing goals but as a singular team with a common goal for a client. Ideally, you will have a professional relationship with the physical therapist to which you are referring a client. Regardless, you will want to be sure that the therapist you refer your client to is extremely professional, results-driven, effective, and knows how to work together with a coach to get the best results. In order for this to occur, the therapist must have confidence that you have an understanding of the body and its musculoskeletal functions.

For example, after assessing the client, the physical therapist might report the following feedback:

- No major orthopedic issues
- Left biceps femoris tightness
- Weak right glute medius

This feedback should come with a list of exercises/movements and stretches to help reduce, and ideally eliminate, the aforementioned issues.

Physical Therapy Direct Access

As of 2014, in most all U.S. states an individual does not need a prescription from a physician to have an initial evaluation from a physical therapist. However, a doctor's prescription may be required for insurance reimbursement. Your client should check with the American Physical Therapy Association (APTA) regarding state-by-state physical therapy laws. If you live outside the U.S., be sure to check the rules and regulations regarding this issue.

❖ SUMMARY

- The body is made up of three classes of levers; a class 3 lever is the most common type of lever in the human body.
- Pronation: medial rotation of a body part
- Supination: external rotation of a body part
- Biomechanics are categorized into *static* and *dynamic.* Static refers to the body at rest, while dynamic biomechanics refers to the body in motion.
- It is more difficult to unlearn a learned skill than to learn a skill for the first time.
- Injuries affect biomechanics.
- Muscle imbalances can cause incorrect movement patterns.
- The *kinetic chain* refers to how the body works together as a whole, and not as separate and individually functioning parts. This means that a breakdown in biomechanical form in one area of the body will likely have consequences in other areas of the body.
- A goniometer is an instrument that is used to measure angles of the body.
- When performing biomechanical assessments, it is important to assess the individual from the following aspects: anterior, posterior, lateral.
- Leg length discrepancies (LLDs) are often functional rather than structural.
- Everyone has muscle imbalances and postural issues. It is the degree of muscle imbalance or postural deviation that is the real issue.
    - If your client presents with what you consider to be a postural issue, it is strongly advised to refer the person to a specialist such as a physical therapist.
- The *overhead squat screen* (OSS) is a comprehensive assessment that dynamically evaluates many areas, including hamstring, back, calf, chest, hip abductors, and adductors.
- When performing assessments, practice is the key to become better and more accurate.
- It is always appropriate to refer your client to a specialist.